Homework 4

Information

Name	UNI
Zhewen Guo	zg2567

Goals

Homework: Display and compare the ERP for target vs. non-target flashes.



Channel ten in the stored variable shows when a signal has flashed and which row and column were flashed. Because the keyboard used is 6 x 6, as seen above, values 1-6 in channel ten identify the row, and values 7-12 identify the columns (so 4 → row 4 and 10 → column 4). For example, if the target character is 'N', it means when channel ten is either 3 (row) or 8 (column), the target character 'N' has flashed.

Each time the row or column of the target character flashes, it will be stored in channel eleven. Combine this information to segment the train1_*group{X}_section{Y}*.mat EEG data into target and non-target flashes.

• Use channels ten and eleven as triggers to cut the EEG data. (1 pt)

• Cut 100ms before onset through 600ms after each flash and separate target and non-target characters. (2 pts)

• Take the average response over trials and display the ERP waveforms for a) target and b) non-target conditions for all channels and the average signal over all channels. (6 pts)

• Do you see a P300 response to the target? (1 pt)

Hint: To find the onset of all flashes in channel ten, you can use the following MATLAB script:

```
Index = find (diff ( trig) > 0 );
```

You will still need to separate target and non-target flashes using channel eleven.

* Channel 11 contains target information only in Copy Spelling, not Free Spelling.

Figure 1. Screenshot of Requirements

Task 1: Cut the EEG Data

EEG data contains 11 channels. The first channel contains runtime information. The following 8 channels show the values of EEG signals from Fz, Cz, P3, Pz, P4, PO7, Oz, and PO8. Channel 10 decodes the position of the flashed signal. The values of Channel 10 range from 1 to 12 and last for 4 sample points when a row or column of letters is flashed. Channel 11 outputs a pulse signal (last for 4 sample points) whenever the target letter flashes.

As shown in Figure 2, target flash appears when channel 11 outputs 1. And when Channel 10 outputs a non-zero value while Channel 11 outputs 0, it means there is a non-target flash. (Full code in Appendix 1or 'trigger_alignment.m')

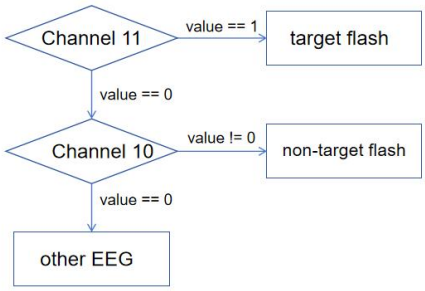


Figure 2. Flowchart of data segmentation

Task 2&3: Capture ERP of target and non-target characters.

After saving data from task 1, we could keep previous 26 sample points (100ms) and subsequent 154 sample points (600ms) of target and non-target flash. (For Task 2: check 'ERP_calculate.m' in folder 'Code' or Appendix 2 for how to detect the onset point and cut the data by Matlab)

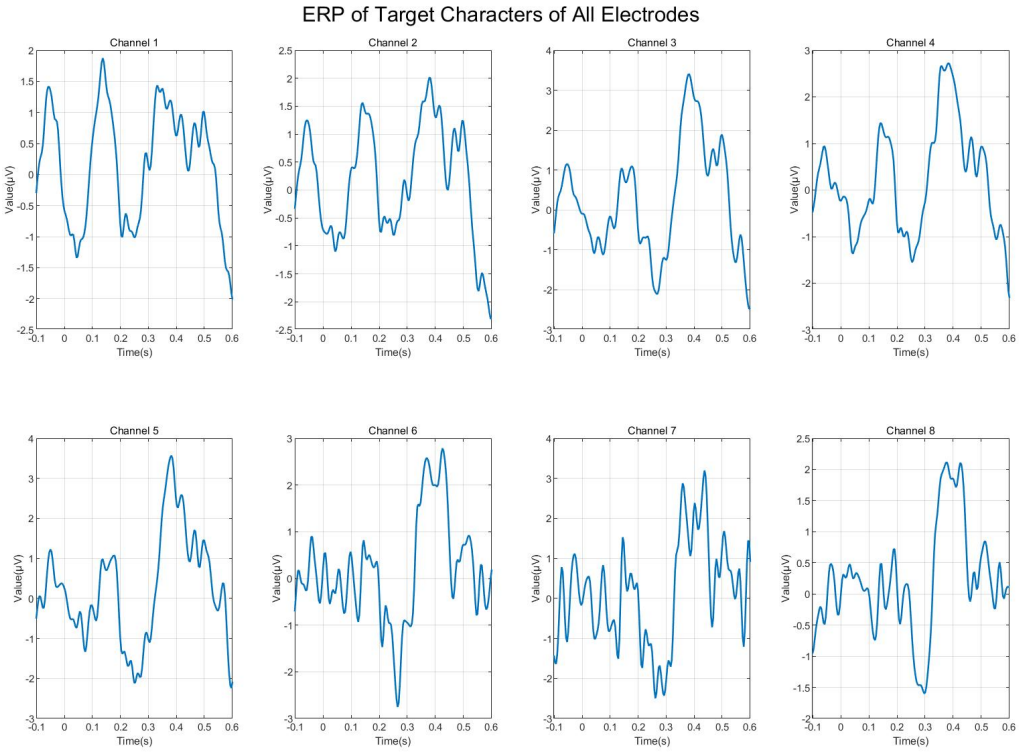


Figure 3. ERP of target character of all electrodes

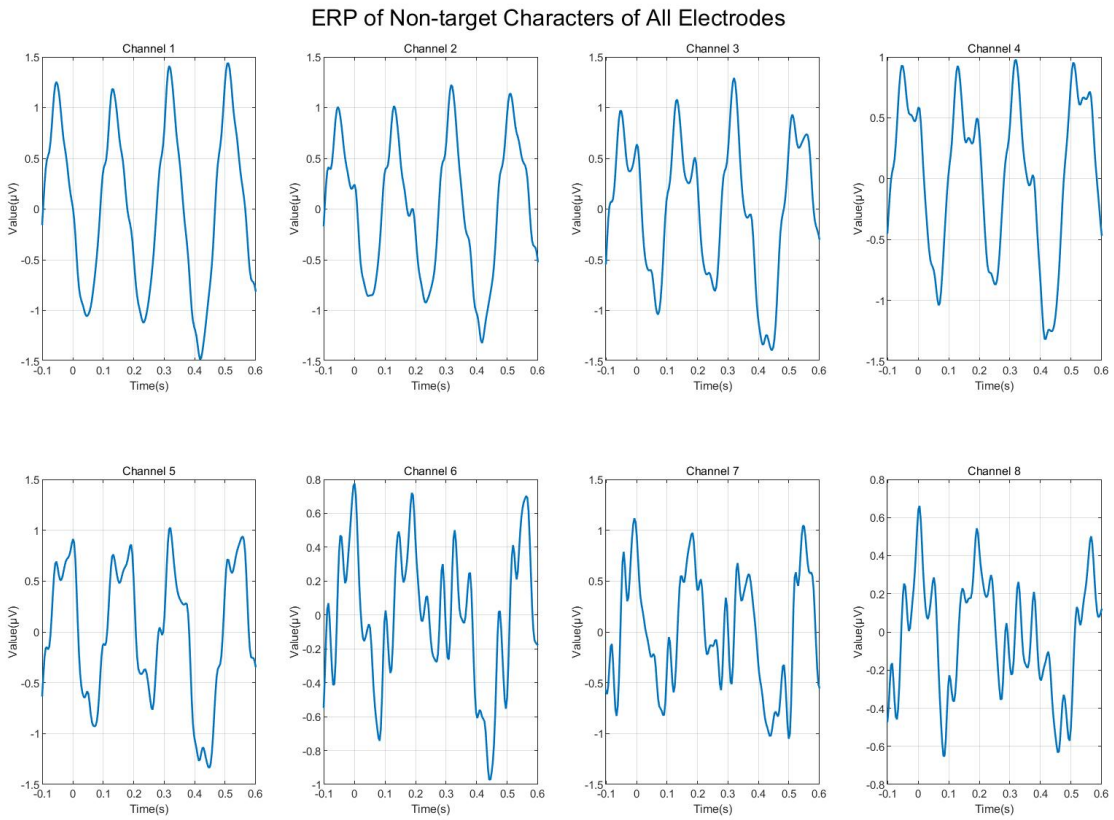


Figure 4. ERP of non-target character of all electrodes

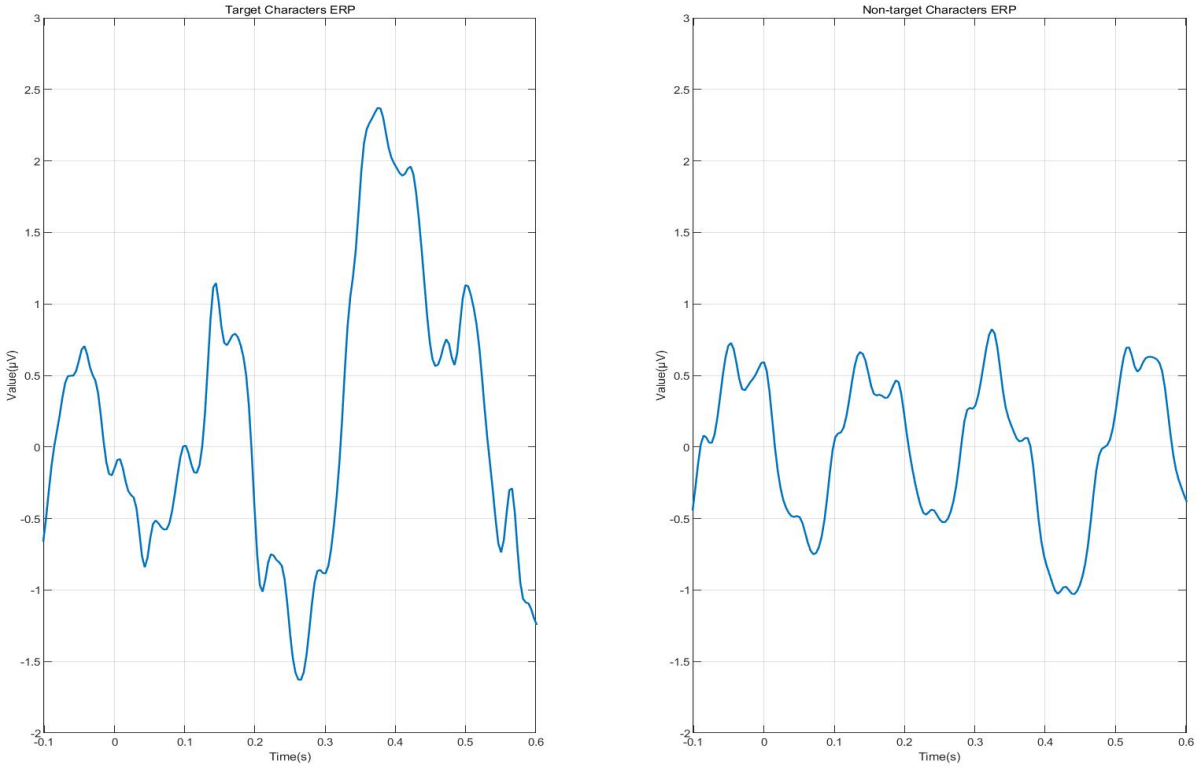


Figure 5. Average ERP signal over all channels

Figure 3 (target flash) and Figure 4 (non-target flash) exhibit the ERP signals of all 8 electrodes. Figure 5 display the average ERP waveforms for target (left) and non-target flash(right) over all channels.

Task 4: P300 Detection

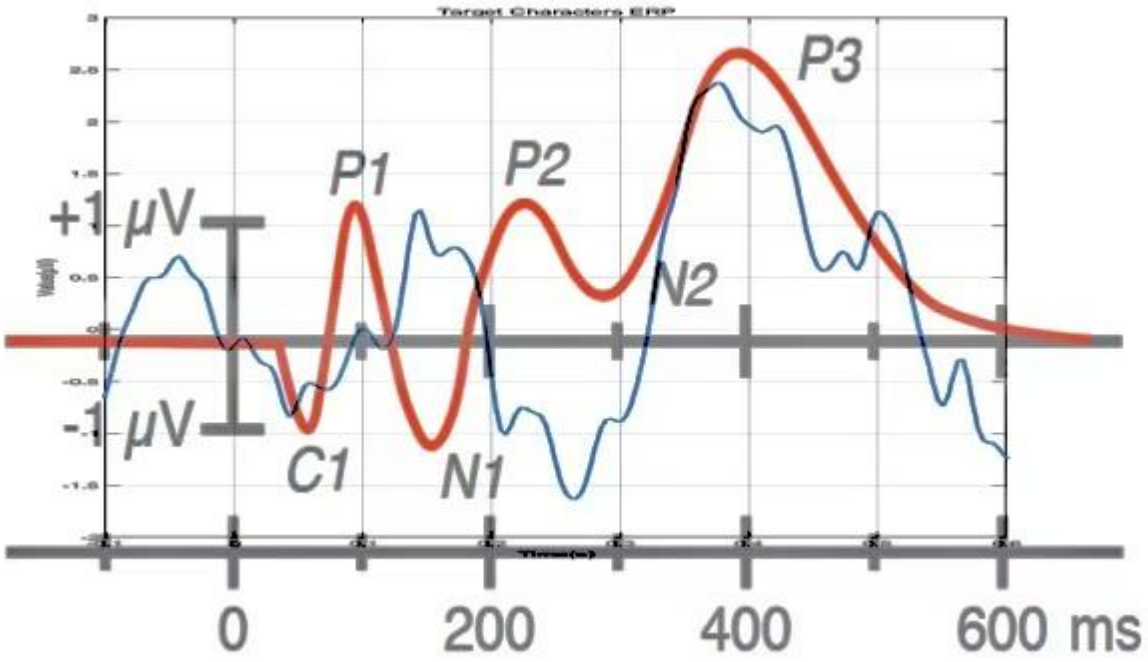


Figure 6. Target flash ERP signal (blue) vs. Standard ERP signal (red)

Figure 6 shows the comparison of the mean ERP (blue) in this experiment and a standard ERP (red). P3 means P300 component. It is evident that there is a apparent P300 component to the target in the average ERP data.

Appendix 1

```
function [m, n] = trigger_alignment(trigger, target)
m = zeros(1, length(trigger)); %non-target
n = zeros(1, length(trigger)); %target
k = 0;
j = 0;
for i = 2:length(trigger)
if target(i) == 1 && target(i-1) == 0
k = k + 1;
n(k) = i;
else
if trigger(i) ~= 0 && trigger(i-1) == 0
j = j + 1;
m(j) = i;
end
end
end
fprintf("find %d target pulse\n", k);
fprintf("find %d non-target pulse\n", j);
n = n(1:k);
m = m(1:j);
end
```

Appendix 2

```
function y = ERP_calculate(trigger, signal)
%signal ==> 8*n
before_sample = 26; % 0.1s * 256 (sample rate)
after_sample = 154; % 0.6s * 256 (sample rate)

y = zeros(size(signal,1),before_sample+after_sample+1);

for i = 1:size(signal,1)
temp_raw = data_capture(trigger, signal(i,:), before_sample, after_sample);
temp_mean = mean(temp_raw,1);
y(i,:) = temp_mean;
end
end
```